

BANARAS HINDU UNIVERSITY

(Detailed Syllabus of different Courses under NEP 2020)

Department of Environment and Sustainable Development

Faculty of Environment and Sustainable Development

Course Title & Code	Environmental Science – IESVA11				
Category of Course ¹	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)				
Credits ² & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	02	00	02	
	Teaching Hours (Total)	30	00	30	
Course Objectives (not more than 100 words)	To provide students with a broad interdisciplinary framework to understand the complex environmental challenges being faced by humanity, and to equip them to live sustainably and in harmony with nature.				
Course Content	The course content is divided into 5 units along with defined hours of teaching for each unit.				
Units	Course Content			Teaching Hours	
I	Human and the Environment Human-Environment Interactions: Historical and current perspectives including the Neolithic period, industrialization, and current human population growth; The development of environmental ethics and the rise of environmentalism.			4	
II	Environment and Natural Resources Components of the Environment: atmosphere, hydrosphere, lithosphere, and biosphere; Natural resources: physical and biological.			7	
III	Ecosystem and Biodiversity Basic concepts of ecosystem; Biotic and abiotic components; Nutrient and energy flow in ecosystems; Overview of terrestrial and aquatic ecosystems; Ecosystem services; Biodiversity: threats, conservation and management.			6	
IV	Environmental Issues Environmental pollution; Global climate change and ozone layer depletion; Hazards, disasters, and their management; Human migration and its environmental impacts; Urbanization and its challenges; Pandemics and communicable diseases in relation to the environment.			7	
V	Environmental Management and Sustainable Development Environmental protection, management and legislation in India; Key international environmental treaties and policies. Sustainable development concepts: Tragedy of commons; UN-Sustainable Development Goals; Environmental education; Nature-friendly lifestyle and nature-based solutions.			6	

¹ SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

² 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Texts / References	• Barucha E (2012) Textbook for Environmental Studies. University Grants Commission, Universities Press. • De AK (2017) Environmental Chemistry. 8th Edition, New Age International Publishers. • Hassenzahl DM, Hager MC, Gift NY, Berg LR, Raven PH (2018) Environment. Wiley. • Harris F (2012) Global Environmental Issues. 2nd Edition, Wiley-Blackwell. • Kellar E. (2021) Introduction to Environmental Geology, 5th Edition, Pearson. • Sharma PD (2018) Ecology and the Environment. Rastogi Publishers. • Singh JS, Singh SP, Gupta SR (2021). Ecology, Environmental Science and Conservation. S. Chand Publishing. • Theodore MK, Theodore L (2024) Introduction to Environmental Management. 2nd Edition, CRC Press, Taylor & Francis Group. • Valdiya K. S. (2004) Geology, Environment and Society, University Press.	
Reports/Magazines	• Down to Earth, Centre for Science and Environment, New Delhi • The State of India's Environment reports, Centre for Science and Environment, New Delhi. • Survey of the Environment, The Hindu • The Global Biodiversity Outlook, UNEP • The Global Environmental Outlook, UNEP	
Learning Outcomes (not more than 100 words)	This value-added course (VAC) is designed to help students: (i) Gain a deep understanding of the human-environment interactions, and the sustainable utilization of natural resources. (ii) Understand the concept of sustainable development and global efforts to achieve these goals. (iii) Embrace a nature-friendly lifestyle that promotes the well-being of both people and the planet.	

Head of the Department